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1 What is WPI

* The Worcester Polytechnic Institute is a private research university in Worcester, Massachusetts, United States. Founded in 1865, WPI was one of the United States' first engineering and technology universities and now has 14 academic departments with over 50 bachelor's, master's, and Ph.D. degree programs.

The FIRST Robotics Competition (FRC) at WPI ([Worcester Polytechnic Institute](#)) refers to WPI's deep involvement in hosting and supporting the national FRC program for high school students, where teams build and compete with industrial-sized robots, fostering STEM skills, teamwork, and leadership, with WPI providing resources like their WARP preseason event and hosting competitions. It's a hands-on experience where students learn engineering, programming, and business, guided by mentors, all while aiming for scholarships and championship events.

2 Git installs

2.1 Create GitHub account

Tutorial : [How to Create a GitHub Account \(For Beginners!\)](#)

[The Ultimate GitHub SSH Key Tutorial \(Fast & Easy\)](#)

Generate ssh key commands: [Generating a new SSH key and adding it to the ssh-agent - GitHub Docs](#)

2.2 Download and install

Tutorial: [Git, GitHub, & GitHub Desktop for beginners](#)

1> Git from [Git](#) (link)

Git from command line interface (GitBash).

Tutorial: [Git and GitHub Tutorial for Beginners](#)

OR

2> “GitHub Desktop Client” from [GitHub Desktop | Simple collaboration from your desktop · GitHub](#) (link)

Alternatively, instead of using Git from command line , GitHub Desktop Client allow us to use Git from graphical interface.

If you have [GitHub Desktop](#) installed, you can use it to clone repositories and not deal with SSH keys.

3 VS code and Git

Tutorial: [How To Use Git In VS Code Like A Pro!](#)

4 Getting Started

[Introduction — FIRST Robotics Competition documentation](#)

5 System Installation

FGR Startup notes (link): [Byting Irish Programming 2023 - HackMD](#)

6 WPI Library

More info: [WPILib](#)

The WPI Robotics Library (WPILib) is the standard software library provided for teams to write code for their FRC® robots. WPILib contains a set of useful classes and subroutines for interfacing with various parts of the FRC control system (such as sensors, motor controllers, and the driver station), as well as an assortment of other utility functions.

6.1 How to install WPILib

Download and Install WPILib from

[WPILib Installation Guide — FIRST Robotics Competition documentation](#)

From the above steps following are installed

WPILib comes with customized VSCode. It not required to uninstall if another VSCode exist until SYNC is disabled.

The WPILib extension provides the FRC® specific functionality related to creating projects and project components, building, and downloading code to the roboRIO and more. You can access the WPILib commands one of two ways:

- By typing "WPILib" into the Command Palette
- By clicking on the WPILib icon in the top right of most windows. This will open the Command Palette with "WPILib" pre-entered

Here is link understand the WPILib's features/WPI Extensions/Commands/Create project etc in VSCode.

Link : [Visual Studio Code Basics and the WPILib Extension — FIRST Robotics Competition documentation](#)

What is Installed?

The Offline Installer installs the following components:

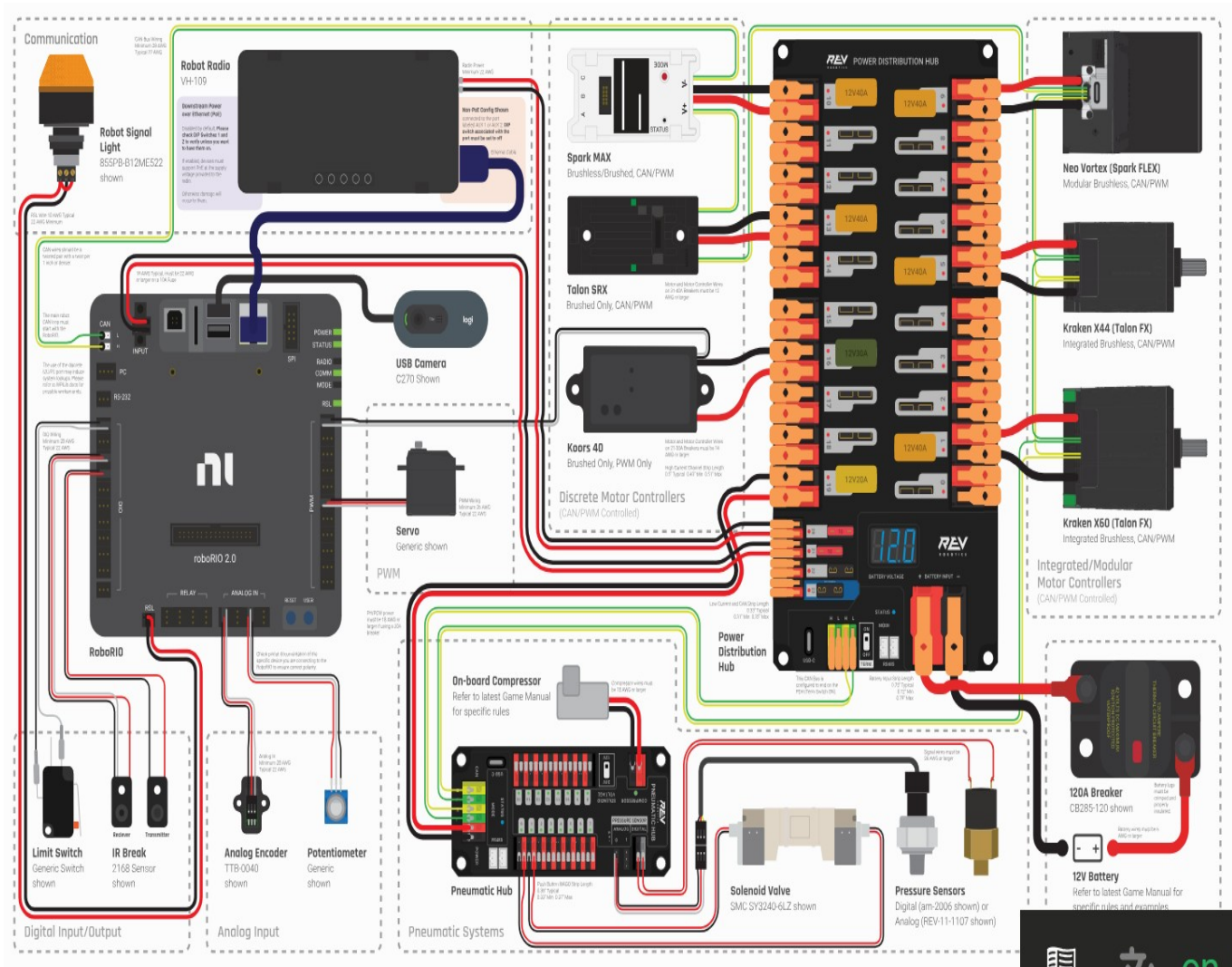
- **Visual Studio Code** - The supported IDE for 2019 and later robot code development. The offline installer sets up a separate copy of VS Code for WPILib development, even if you already have VS Code on your machine. This is done because some of the settings that make the WPILib setup work may break existing workflows if you use VS Code for other projects.
- **C++ Compiler** - The toolchains for building C++ code for the roboRIO
- **Gradle** - The specific version of Gradle used for building/deploying C++ or Java robot code
- **Java JDK/JRE** - A specific version of the Java JDK/JRE that is used to build Java robot code and to run any of the Java based Tools (Dashboards, etc.). This exists side by side with any existing JDK installs and does not overwrite the JAVA_HOME variable
- **WPILib Tools** - [SmartDashboard](#), [Shuffleboard](#), [RobotBuilder](#), [OutlineViewer](#), [PathWeaver](#), [Glass](#), [SysId](#), [Data Log Tool](#), [roboRIO Team Number Setter](#), [AdvantageScope](#), [Elastic](#), [WPICal](#)
- **WPILib Dependencies** - OpenCV, etc.
- **VS Code Extensions** - WPILib and Java/C++/Python extensions for robot code development in VS Code
- **Documentation** - Offline copies of this [frc-docs](#) documentation and Java/C++/Python APIs

Note

It's not recommended to use the VS Code Backup and Settings Sync feature to sync settings between a regular VS Code installation and the FRC VS Code installation. This could lead to either installation being broken.

7 Hardware introduction

Hardware Component Overview — FIRST Robotics Competition documentation



8 Software components for 2025

List of software components are listed here - [Software Component Overview — FIRST Robotics Competition documentation](#)

Installing guide of software - [Step 2: Installing Software — FIRST Robotics Competition documentation](#)

8.1 Download and install Gaming tool

[Installing the FRC Game Tools — FIRST Robotics Competition documentation](#)

The FRC® Game Tools contains the following software components:

- LabVIEW Update
- FRC Driver Station
- FRC roboRIO Imaging Tool and Images

8.2 FRC Driver station

Understand more on Driver station here

[FRC Driver Station Powered by NI LabVIEW — FIRST Robotics Competition documentation](#)

8.3 VSCode that comes with WpiLib

8.4 LabView (optional)

- [What is NI LabVIEW?](#)
- [Basics of LabVIEW Programming Environment Tutorial](#)
- [Beginners LabVIEW Tutorial 1: Getting Started with LabVIEW](#)

8.5 Python (optional)

Install python 3.13 from microsoft store and install RobotPy package

`($python3 -m pip install robotpy)`

[Python Installation Guide — FIRST Robotics Competition documentation](#)

9 Robo code 2025

Clone : [FRC7660/robot-2025](https://github.com/FRC7660/robot-2025)

<path>\$ git clone <https://github.com/FRC7660/robot-2025.git>

10 Imaging RoboRio and setting Team number

[Imaging your roboRIO 2 — FIRST Robotics Competition documentation](#)

11 Building and Deploying Robot Code

For more details on how to build and deploy the code on RoboRio, Check this link

[Building and Deploying Robot Code — FIRST Robotics Competition documentation](#)

12 Programming your Robot

[Creating your Test Drivetrain Program \(Java/C++/Python\) — FIRST Robotics Competition documentation](#)

13 Running your Test Program

[Running your Test Program — FIRST Robotics Competition documentation](#)